Current	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
629 - matmul_name	64, 64, 1	24.74 s, 15.84 ms	34,814,815	0 - NVIDIA GeForce RTX 4050 Laptop GPU - 1.87 GHz		[3012] matmul_name.exe	

Summary | Baseline | Source | Contact | Comments | Raw | Session | Compare | Tools | Open | Export

GPU Speed Of Light Throughput

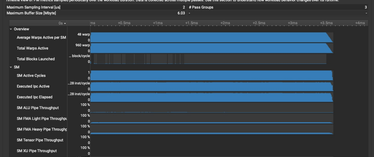
GPU Speed Of Light Throughput — GPU Throughput Chart

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for individual sub-metrics of Compute and Memory to clearly identify the bottleneck. High level overview of the utilization of compute and memory resources of the GPU presented as a roofline chart.

Compute [%]	98.38	Duration [ms]	15.84
Memory Throughput [%]	95.37	Elapsed Cycles [cycle]	43,918,025
L1/TEX Cache Throughput [%]	98.39	Max Active Cycles [cycle]	68,14,626,20
L2 Cache Throughput [%]	95.37	Frequency [GHz]	1.87
DRAM Throughput [%]	1.20	DRAM Frequency [GHz]	7.24

High Throughput This workload is utilizing greater than 60.0% of the available compute or memory throughput of the device. To further improve performance, work will likely need to be shifted from the most utilized to another unit. Start by analyzing workloads in the Compute Workload Analysis section.

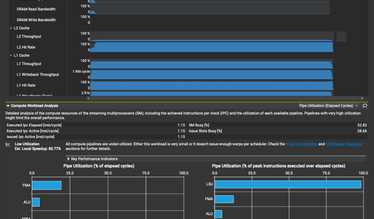
Roofline Analysis The ratio of peak float (fp32) to double (FP64) performance on this device is 64:1. The workload achieved 6% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Roofline chart for more details on roofline analysis.



PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us]	2	Pass Groups	1
Maximum Buffer Size [MiBytes]	6,03		



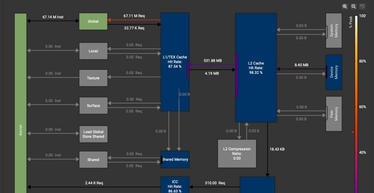
Compute Workload Analysis

Pipe Utilization (Elapsed Cycles)

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	1.11	SM Busy [%]	32.82
Executed Ipc Active [inst/cycle]	1.13	Issue Slots Busy [%]	28.44
Issued Ipc Active [inst/cycle]	1.14		

Low Utilization All compute pipelines are under-utilized. Either this workload is very small or it doesn't issue enough warps per scheduler. Check the Launch Statistics and Scheduler Statistics sections for further details.



Memory Workload Analysis

Memory Chart

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between the units (Max Bandwidth), or by reaching the maximum throughput being issued across the various memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [GBytes/s]	2.30	Mem Busy [%]	73.61
L1/TEX Hit Rate [%]	87.54	Max Bandwidth [%]	98.46
L2 Hit Rate [%]	98.32	Mem Pipes Busy [%]	98.34
Mem Footprint [MBytes]	1,024	Max Shared Memory Requests	1
L2 Compression Ratio	0	Shared Memory Spilling Requests	0
L2 Compression Success Rate [%]	0	Local Memory Spilling Request Overhead [%]	0
L2 Persisting Size [MByte]	4,72	Shared Memory Spilling Request Overhead [%]	0
L2 Sector Promotion Misses [%]	0		

L1/TEX Global Load Access Pattern **Est. Speedup: 43.53%** The memory access pattern for global loads from L1/TEX might be not optimal. On average, only 18.0 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the Memory Chart section for uncoalesced global loads.

Memory Chart
Values
Transfer Size
Inactivity
Staged Out
Device
Global
Local
Texture
Surface
Local Global Store Shared
Shared
L1/TEX Cache
Shared Memory
L2 Compression Fails
ICC
MC
ICC Hit Rate
L2 Cache
Hit Rate
98.32 %
Mem Pipes Busy
Mem Busy
Max Bandwidth
Mem Busy
0.00 B
0.00 B
0.00 B